

Permutations

Counting problems ask you to count the possible ways that something could happen. Sometimes the order of events matters and sometimes it doesn't. When order matters, you are counting the number of **permutations**.

In real life, the word *combination* is often used when really it is a *permutation*. When you use a combination lock, you have to enter the numbers in a certain order. A combination lock should really be called a permutation lock.



You will learn more about **combinations** in the next lesson. For now, remember that in a permutation problem, the order of the events or items matters.

Example 1

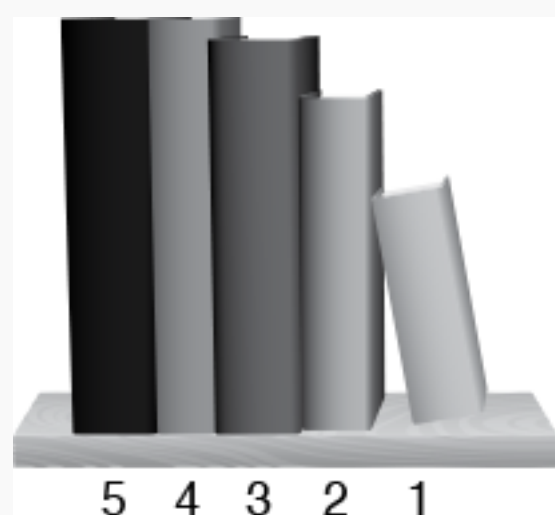
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In how many different ways can a student arrange five different books on a shelf?

Step 1

Count the number of choices the student has for each space on the shelf.

There are 5 choices for the first book. After a book is placed in that position, there are 4 choices for the second book. After each book is placed, there is 1 fewer choice. Notice how the choices count down after each book is placed.



Step 2

Multiply the choices: $5 \times 4 \times 3 \times 2 \times 1 = 120$

There are **120 different ways** the books can be arranged on the shelf.

To solve any permutation problem, simply multiply the choices. Study how this basic idea can be used to solve a more complex problem.

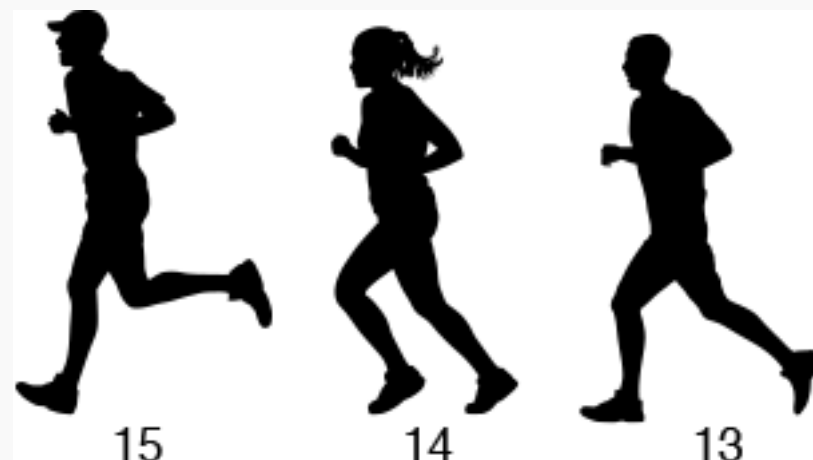
Example 2

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Fifteen runners are in a race. How many different finishes can there be for 1st, 2nd, and 3rd place?

Step 1

Count the number of choices. There are 15 runners who could finish in 1st place, 14 choices for 2nd place, and 13 choices for 3rd place. Remember, once a runner finishes 1st, that runner cannot also finish in 2nd or 3rd place.



Step 2

Multiply the choices: $15 \times 14 \times 13 = 2,730$

There are **2,730 different ways** the runners could finish 1st, 2nd, and 3rd.

Will the choices always count down? Not always.

Example 3

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A bike combination lock has three dials, each with 10 numbers. To open the lock, you turn the dials. A number can be used *more than once*. How many permutations are possible?

There are 10 choices for each dial: $10 \times 10 \times 10 = \mathbf{1,000}$.



Key Terms

combination

permutation

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